

University of Utah

Fall 2025

MATH 3220-002

Final Exam Questions

Instructor: Alp Uzman

December 12, 2025, 8:00 AM - 10:00 AM

Surname:

First Name:

uNID:

KEY

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1. [20 points] Let $v_\bullet$ be a sequence of vectors in $\mathbb{R}^5$ and $L_\bullet$ be a sequence of 3×5 matrices with real entries. Consider the following statements:

A: $v_\bullet$ is a bounded sequence of vectors.

B: $v_\bullet$ converges to the zero vector in $\mathbb{R}^5$.

C: $L_\bullet$ is a sequence of matrices bounded with respect to the operator norm.

D: $L_\bullet$ is a sequence of matrices that converges to the zero matrix with respect to the operator norm.

E: $\lim_{k \rightarrow \infty} L_k v_k = 0$ in $\mathbb{R}^3$.

F: $\lim_{\ell \rightarrow \infty} \lim_{k \rightarrow \infty} L_\ell v_k = 0$ in $\mathbb{R}^3$.

- (a) [5 points] **Prove or disprove the following statement:**

If **B**, then **A**.

$$v_n \rightarrow 0 \Rightarrow \exists N, \forall n \geq N, |v_n| < 1$$

$$\Rightarrow \forall n \geq 0: |v_n| \leq \max \{1, |v_0|, |v_1|, \dots, |v_n|\} < \infty \quad \square$$

- (b) [5 points] **Prove or disprove the following statement:**

If **B** and **C**, then **F**.

$$|L_\ell v_k| \leq \underbrace{\|L_\ell\|_{op}}_{< \infty} \underbrace{|v_k|}_{\rightarrow 0} \xrightarrow{k \rightarrow \infty} 0$$

$$\Rightarrow \lim_{\ell} \lim_k L_\ell v_k = \lim_{\ell} 0 = 0. \quad \square$$

- (c) [5 points] Prove or disprove the following statement:
If A and D, then E.

$$|L_k v_k| \leq \underbrace{|L_k|}_{\rightarrow 0} \underbrace{|v_k|}_{< \infty} \rightarrow 0.$$

□.

- (d) [5 points] Prove or disprove the following statement:
If E, then F.

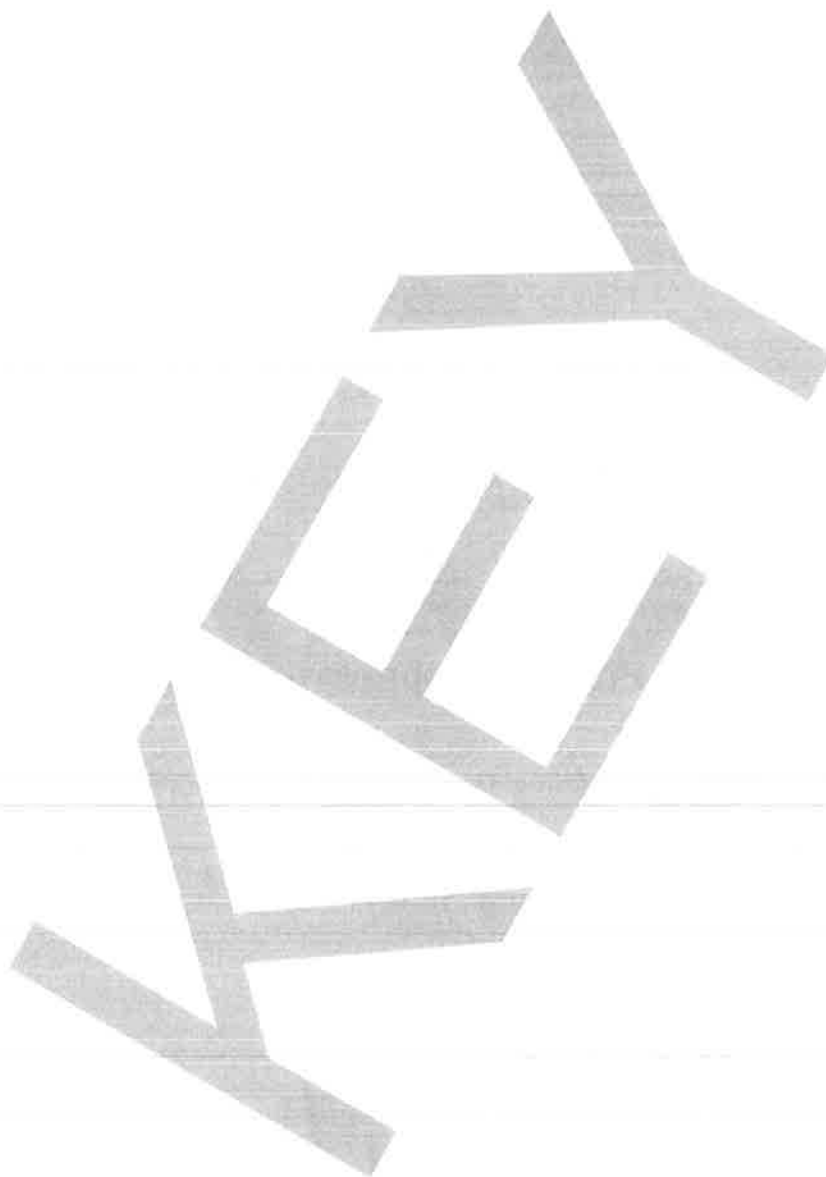
$$L_\ell = \frac{1}{\ell^2} \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad v_k = k \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow L_\ell v_k = \frac{k}{\ell^2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \lim_{k \rightarrow \infty} L_k v_k = \lim_{k \rightarrow \infty} \frac{1}{k} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \textcircled{E} \text{ ok.}$$

$$\lim_{\ell \rightarrow \infty} \lim_{k \rightarrow \infty} L_\ell v_k = \lim_{\ell \rightarrow \infty} \underbrace{\lim_{k \rightarrow \infty} \frac{k}{\ell^2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}}_{\text{DNE}} = \text{DNE} \Rightarrow \textcircled{F} \text{ not ok.}$$

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2. [20 points] There is nothing to prove in this problem. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$(x, y) \mapsto e^{xy}.$$

Perform the following tasks:

- (a) [10 points] Compute the derivative of f at $(0, 0)$.

$$\begin{aligned} f'(x, y) &= \left(\frac{\partial f}{\partial x}(x, y) \quad \frac{\partial f}{\partial y}(x, y) \right) \\ &= \left(e^{xy} y \quad e^{xy} x \right) \\ &= f(x, y) \cdot (y \ x) \\ \Rightarrow f'(0, 0) &= f(0, 0) \cdot (0 \ 0) = \boxed{(0 \ 0)}. \end{aligned}$$

- (b) [5 points] Compute the directional derivative of f at $(0, 0)$ in the direction of $(1, -1)$.

$$D_{(1, -1)} f(0, 0) = f'(0, 0) (1, -1) = (0, 0) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \boxed{0}.$$

(c) [5 points] Compute the order 2 Taylor polynomial of f at $(0,0)$.

$$f''(x,y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2}(x,y) & \frac{\partial^2 f}{\partial x \partial y}(x,y) \\ \frac{\partial^2 f}{\partial y \partial x}(x,y) & \frac{\partial^2 f}{\partial y^2}(x,y) \end{pmatrix}$$

2
1
0

$$\begin{array}{c} \uparrow \\ f \cdot x^2 \\ \uparrow \\ f \cdot x \rightarrow f(xy+1) \\ \uparrow \quad \uparrow \\ f \xrightarrow{\frac{\partial}{\partial y}} f \cdot y \xrightarrow{\frac{\partial}{\partial x}} f \cdot y^2 \rightarrow f \cdot y^3 \rightarrow \dots \end{array}$$

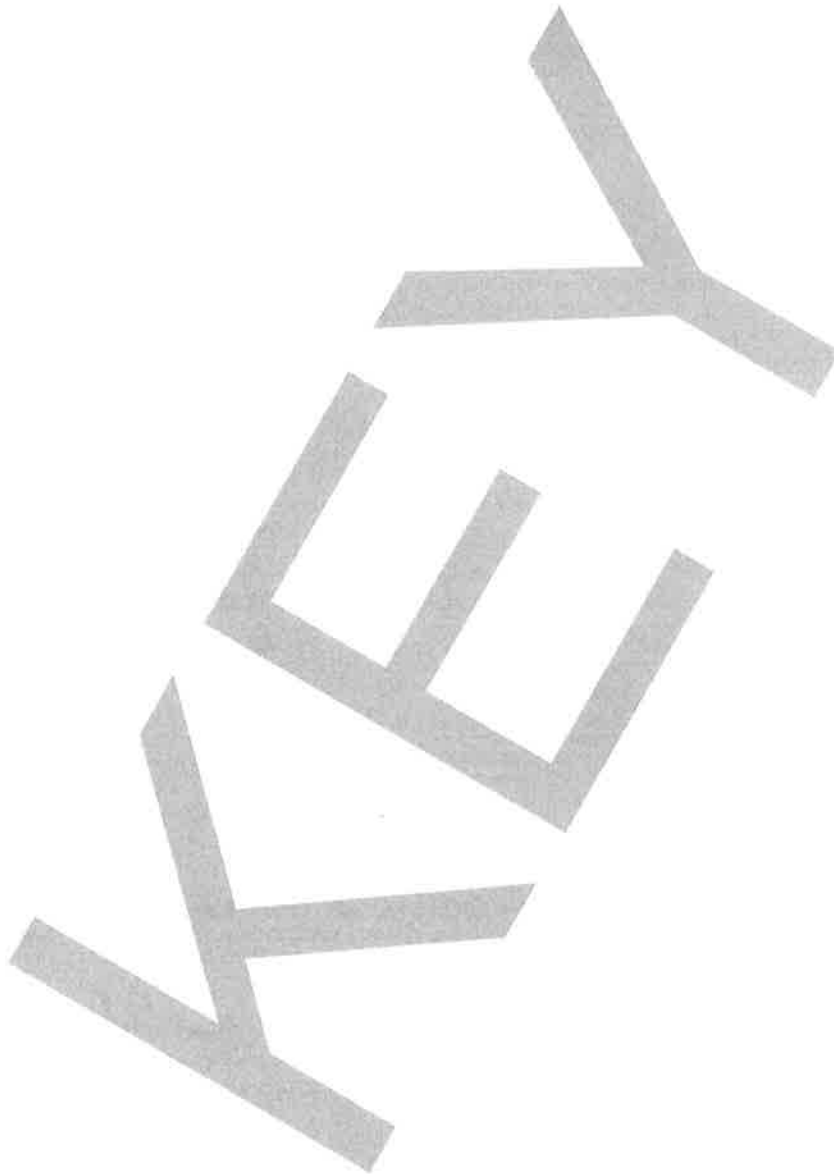
$$\frac{\partial}{\partial y}(f \cdot y) = \frac{\partial f}{\partial y} \cdot y + f = f \cdot xy + f = f(xy+1)$$

$$\Rightarrow f''(0,0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Also $f(0,0) = e^0 = 1 \neq 0$

$$\begin{aligned} \Rightarrow T_2(f; (0,0))(x,y) &= f(0,0) + f'(0,0)(x,y) + \frac{1}{2} f''(0,0)(x,y)^2 \\ &= 1 + \frac{1}{2} \begin{pmatrix} x \\ y \end{pmatrix}^T \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \boxed{1 + xy} \end{aligned}$$

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3. [20 points] Let $f : \mathbb{R}^5 \rightarrow \mathbb{R}^4$ be a continuously differentiable function. Consider the following statements:

X: The rank of $f'(0)$ is 4.

Y: There is a continuously differentiable $g : \mathbb{R}^5 \rightarrow \mathbb{R}^1$ such that the function $\mathbb{R}^5 \rightarrow \mathbb{R}^5$ defined by $y \mapsto (f(y), g(y))$ is a C^1 diffeomorphism near the origin.

- (a) [10 points] Prove or disprove the following statement:
If X, then Y.

rank $(f'(0)) = 4 \Leftrightarrow$ Relative to some
bases, $f'(0) = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$

Let ~~f~~ $g : \mathbb{R}^5 \rightarrow \mathbb{R}$, be such that
relative to the same bases,

$$g(x_1, x_2, x_3, x_4, x_5) = x_5 \quad (g \text{ linear})$$

$$\Rightarrow g'(0) = (0 \ 0 \ 0 \ 0 \ 1).$$

$$\Rightarrow (f, g)'(0) = I \text{ rel. } \underline{\text{same bases.}}$$

$\Rightarrow$ Inv. T.T. $\Rightarrow f, g$ local C^1 diffeo. near
0.

(b) [10 points] Prove or disprove the following statement:
If Y, then X.

(f, g) local diffeomorphism near 0

$\Rightarrow (f, g)'(0)$ inv.

$$\Rightarrow (f, g)'(0) = \begin{pmatrix} f'(0) \\ g'(0) \end{pmatrix}$$

$f'(0) : \mathbb{R}^5 \rightarrow \mathbb{R}^4 \Rightarrow$ By Rank Nullity,
 $0 \leq \text{rank}(f'(0)) \leq 4$

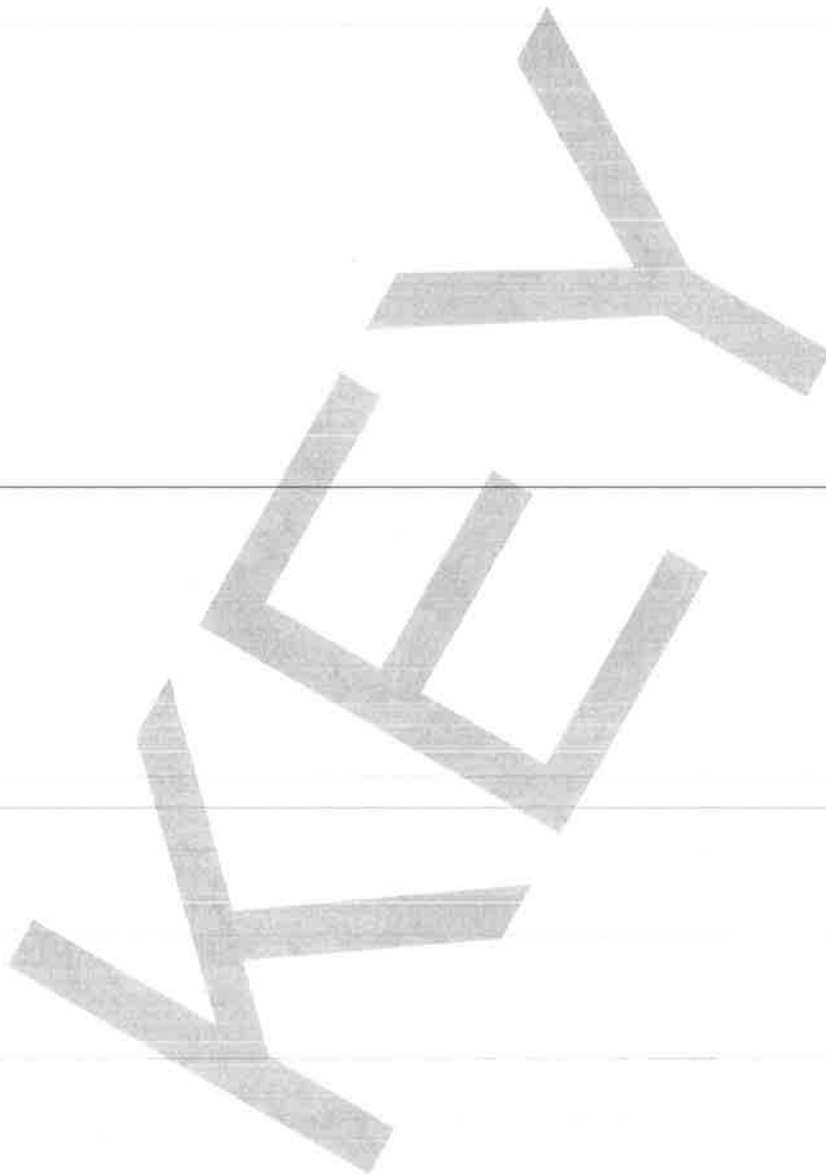
$g'(0) : \mathbb{R}^5 \rightarrow \mathbb{R} \Rightarrow$ By Rank Nullity,
 $0 \leq \text{rank}(g'(0)) \leq 1$

$(f, g)'(0)$ inv $\Rightarrow$ full rank

$\Rightarrow 5$ pivots $\Rightarrow \text{rank}(f'(0)) = 4$.
rows.

$\square$.

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4. [20 points] There is nothing to prove in this problem. Put

- $M = \{(x, y) \in \mathbb{R}^2 \mid (x, y) \neq (0, 0)\}$,
- $f = xy \in C^\infty\text{-Form}^0(M)$,
- $\omega = xdy \in C^\infty\text{-Form}^1(M)$,
- $\gamma \in C^\infty(\mathbb{R}; M)$, $t \mapsto (\cos(t), \sin(t))$,
- $\alpha = 0$, $\beta = -4\pi$.

Perform the following tasks:

(a) [5 points] Compute df .

$$df = d(xy) = \boxed{y dx + x dy.}$$

(b) [5 points] Compute $d\omega$.

$$d\omega = d(xdy) = \boxed{dx \wedge dy.}$$

(c) [5 points] Solve $dg = \omega$ for g a C^1 differential 0-form on M .
(It is possible this equation has no solution.)

ω not closed $\Rightarrow \omega$ not exact

$$\Rightarrow \boxed{\text{No } g : dg = \omega.}$$

(d) [5 points] Compute $\left\langle \frac{1}{3} [\gamma; \beta \leftarrow \alpha] \mid -df \right\rangle$.

Only because
I don't want to
terrify you by perds
↓ shilla.

~~$$[\gamma; \beta \leftarrow \alpha] | \omega \rangle = \int_{\alpha}^{\beta} \omega(\gamma(t)) \gamma'(t) dt$$~~

~~$$= \int_0^{-4\pi} \left(\frac{0}{\cos(t)} \right) \left(\frac{-\sin(t)}{\cos(t)} \right) dt = - \int_{-4\pi}^0 \cos^*(t) dt$$~~

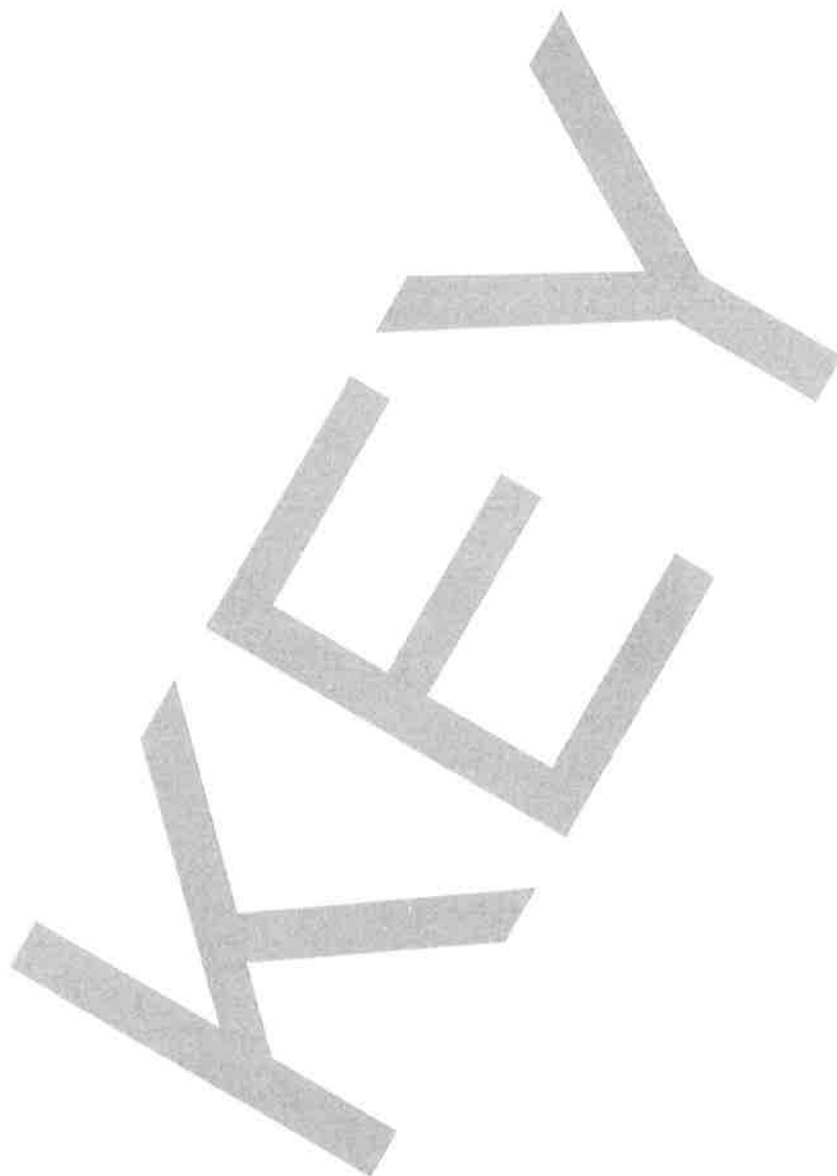
$$\left\langle \frac{1}{3} [\gamma; \beta \leftarrow \alpha] \mid -df \right\rangle$$

$$= -\frac{1}{3} \langle [\gamma; \beta \leftarrow \alpha] \mid df \rangle$$

$$\stackrel{z}{=} -\frac{1}{3} \langle \partial \cdot [\gamma; \beta \leftarrow \alpha] \mid f \rangle$$

$$\boxed{FTC} \quad = -\frac{1}{3} \underbrace{\langle \partial \cdot [\gamma(p)] - [\gamma(q)] \mid f \rangle}_{=0} = \boxed{0}$$

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5. [15 points] Let $\xi = (\xi_1, \xi_2, \xi_3) \in \mathbb{R}^3$ be nonzero and consider $Q_\xi : \mathbb{R}^2 \rightarrow \mathbb{R}, (x, y) \mapsto \xi_1 x^2 + \xi_2 xy + \xi_3 y^2$. Define

$$M_\xi = \{(x, y) \in \mathbb{R}^2 \mid Q_\xi(x, y) \neq 0\}.$$

- (a) [5 points] Prove or disprove the following statement:
 M_ξ is a nonempty open subset.

Open: $M_\xi = \mathbb{R}^2 \setminus Q_\xi^{-1}(0)$

Q_ξ poly $\Rightarrow$ cont $\Rightarrow$ preimages of closed sets are closed
 $\Rightarrow Q_\xi^{-1}(0)$ closed
 $\Rightarrow M_\xi$ open $\checkmark$.

Nonempty: If $M_\xi = \emptyset$, $\forall (x, y) \in \mathbb{R}^2$:

$Q_\xi(x, y) = 0$. $\forall \xi \in \mathbb{R}^3$, $x \mapsto Q_\xi(x, y)$ is a degree ≤ 2 poly, but by the quadratic formula has at most 2 real zeros.

Recall: We're given Q_ξ is not the zero poly.

(b) [5 points] Prove or disprove the following statement:

On M_ξ , the differential 1-form $\omega = xdy - ydx$ is closed.

$$d\omega = d(xdy - ydx)$$

$$= dx \wedge dy - dy \wedge dx = 2 dx \wedge dy \neq 0.$$

□

(c) [5 points] **Prove or disprove the following statement:**

On M_ξ , the differential 1-form $\frac{\omega}{Q_\xi}$, where $\omega = xdy - ydx$, is closed.

$$d\left(\frac{\omega}{Q_\xi}\right) = d\left(\frac{x}{Q_\xi} dy - \frac{y}{Q_\xi} dx\right)$$

$$= d\left(\frac{x}{Q_\xi}\right) \wedge dy - d\left(\frac{y}{Q_\xi}\right) \wedge dx$$

$$= \frac{\partial}{\partial x} \left(\frac{x}{Q_\xi}\right) dx \wedge dy - \frac{\partial}{\partial y} \left(\frac{y}{Q_\xi}\right) dy \wedge dx$$

$$= \left[\frac{\partial}{\partial x} \left(\frac{x}{Q_\xi}\right) + \frac{\partial}{\partial y} \left(\frac{y}{Q_\xi}\right) \right] dx \wedge dy$$

$$= \left(Q_\xi - x \frac{\partial Q_\xi}{\partial x} \right) + \left(Q_\xi - y \frac{\partial Q_\xi}{\partial y} \right)$$

$$dx \wedge dy = 0.$$

$$Q_\xi^2$$

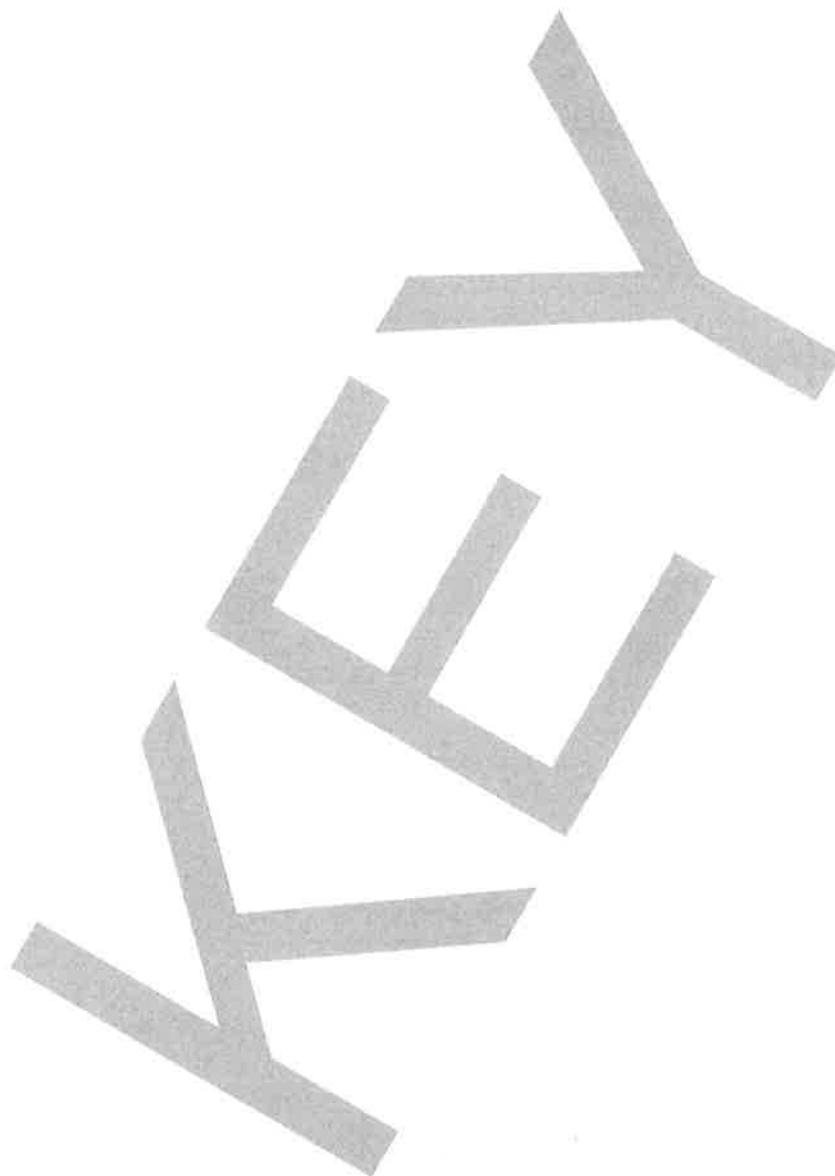
$$\frac{\partial Q_\xi}{\partial x} = 2\frac{y}{\xi}x + \frac{y}{\xi}, \quad \frac{\partial Q_\xi}{\partial y} = \frac{y}{\xi}x + 2\frac{y}{\xi}y$$

$$\Rightarrow 2Q_\xi - x(2\frac{y}{\xi}x + \frac{y}{\xi}) - y(\frac{y}{\xi}x + 2\frac{y}{\xi}y)$$

$$= 2\frac{y}{\xi}x^2 + 2\frac{y}{\xi}xy + 2\frac{y}{\xi}y^2 - 2\frac{y}{\xi}x^2 - \frac{y}{\xi}xy - \frac{y}{\xi}xy - 2\frac{y}{\xi}y^2 = 0$$

□.

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Prove or disprove

6. [3 points] Let $M \subseteq \mathbb{R}^n$ be a nonempty open subset, $x \in M$. A point $y \in M$ is defined to be visible from x if

$$\forall t \in [0, 1] : (1-t)x + ty \in M.$$

(In words, y is visible from x if the line segment connecting x to y is completely within M .)

Then the collection of points in M visible from x is a nonempty path-connected open subset.

Nonempty: $x \in \text{vis}(x)$. $\square$

Path con: $y, z \in \text{vis}(x) \Rightarrow y \rightarrow x \rightarrow z$ path in M . $\square$



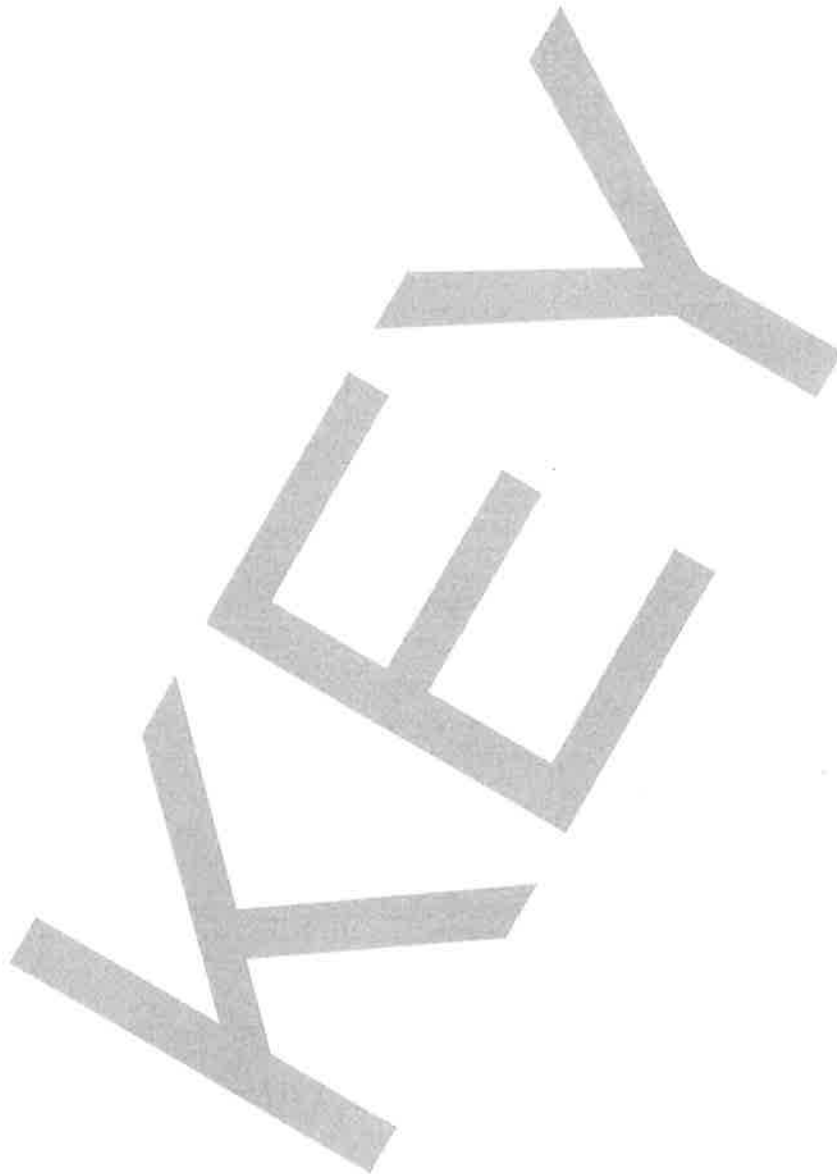
Open: Say $\text{vis}(x)$ is not open

$\Rightarrow \neg [\forall y \in \text{vis}(x), \exists \delta, \forall z : d(z, y) < \delta \Rightarrow z \in \text{vis}(x)]$
 $\equiv \exists y_0 \in \text{vis}(x), \exists y_1 : y_1 \rightarrow y_0, y_1 \notin \text{vis}(x).$

$\Rightarrow \exists t_n \in [0, 1], \forall n : (1-t_n)x + t_n y_1 \notin M.$

$[0, 1]$ compact $\Rightarrow$ BW up to a subseq., $t_n \rightarrow t_\infty \in [0, 1]$,
 By continuity of LA, $(1-t_n)x + t_n y_1 \rightarrow (1-t_\infty)x + t_\infty y_1$.
 Since $y_0 \in \text{vis}(x)$, this limit is in M . M is open
 here for large n , $(1-t_n)x + t_n y_1 \in M$, contradiction. $\square$

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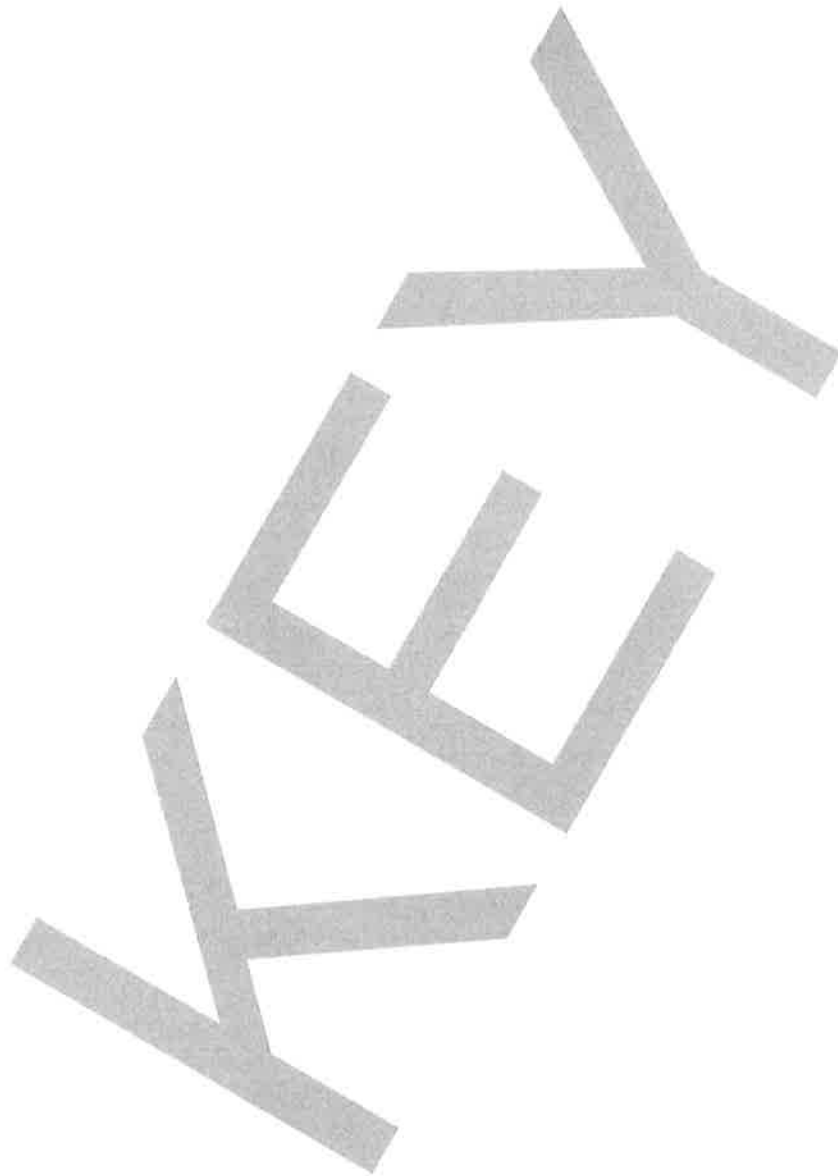
7. [2 points] Let $M \subseteq E$ be an open subset of a normed space, ω be a C^0 differential 1-form on M , $[\gamma; \beta \leftarrow \alpha]$ be a C^1 1-cell in M with $\alpha \leq \beta$. If ω is bounded as a function from M to $\mathcal{BL}(E; \mathbb{R})$, then

$$|\langle [\gamma; \beta \leftarrow \alpha] | \omega \rangle| \leq \text{len}(\gamma|_{[\alpha, \beta]}) \|\omega\|_{C^0}.$$

$$\begin{aligned}
 |\langle [\gamma; \beta \leftarrow \alpha] | \omega \rangle| &= \left| \int_{\alpha}^{\beta} \omega(\gamma(t)) \gamma'(t) dt \right| \\
 &\leq \int_{\alpha}^{\beta} |\omega(\gamma(t)) (\gamma'(t))| dt \\
 &\leq \int_{\alpha}^{\beta} \|\omega(\gamma(t))\|_{\text{op}} \|\gamma'(t)\|_E dt \\
 &\leq \|\omega\|_{C^0} \cdot \int_{\alpha}^{\beta} \|\gamma'(t)\|_E dt \\
 &= \|\omega\|_{C^0} \cdot \text{length}(\gamma|_{[\alpha, \beta]}).
 \end{aligned}$$

□.

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